THE SEVEN- AND EIGHT-COLOR CASES OF AN ERDŐS–GYÁRFÁS BALANCED-COLORING PROBLEM

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Abstract. We record computer-assisted proofs of two fixed cases of a problem of Erdős and Gyárfás. Every seven-coloring of the edges of K_{50} has eight vertices whose induced edges omit a color, and every eight-coloring of the edges of K_{65} has nine vertices whose induced edges omit a color. The seven-color proof uses a direct exhaustive lemma on weighted graphs with seven vertices. The eight-color proof uses a human full-color extremal argument together with 862 independently reconstructed and LRAT-checked finite instances. The accompanying repository contains the source, semantic verifiers, manifests, replay script, and proof artifacts. These results are being circulated for independent checking and have not yet received external mathematical review. Neither result settles the problem for arbitrary r.

1. Introduction

Erdős and Gyárfás asked whether the following holds for every integer $r \geq 3$: whenever the edges of K_{r^2+1} are colored with r colors, some $r+1$ vertices induce a complete graph on which at least one color is absent [5]. The question is also recorded as Erdős Problem 617 [2].

This paper concerns the next two fixed parameters after the previously known and separately treated cases.

Theorem 1.1. *Let $r \in \{7, 8\}$. For every map*

$$\chi : E(K_{r^2+1}) \longrightarrow [r],$$

there is a set $S \subseteq V(K_{r^2+1})$ with $|S| = r+1$ such that

$$\chi(E(K_{r^2+1}[S])) \neq [r].$$

The two parts of Theorem 1.1 are proved by different finite inputs. The $r = 7$ finite lemma is a direct enumeration of 12,618,770 weighted cases, reconstructed a second way using unlabelled graphs and automorphism orders. It does not use SAT. The $r = 8$ proof reduces one branch to a single formula and another to 861 core–shell formulas. Independent semantic checkers reconstruct the formulas, and an LRAT checker verifies their unsatisfiability.

Remark 1.2 (Scope). Theorem 1.1 is a pair of fixed-parameter results. It does not prove Erdős Problem 617 for every r. It does not prove the fixed case $r = 9$, give a Lean formalization, or claim external review. The certificate archives are not embedded in this manuscript.

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2. Shared setup

Assume that every $(r+1)$-set sees all r colors. For a fixed color i, let G_i be the graph formed by its edges. Then

$$\alpha(G_i) \le r. \tag{1}$$

Every $(r+1)$-set contains at least one edge of each of the other $r-1$ colors, so

$$e(G_i[S]) \le \binom{r+1}{2} - (r-1) = \binom{r}{2} + 1 \qquad \left(S \in \binom{V(G_i)}{r+1}\right). \tag{2}$$

For $n = ar + b$, where $0 \le b < r$, put

$$p_r(n) = r\binom{a}{2} + ab. \tag{3}$$

This is the minimum number of edges in an n-vertex graph with independence number at most r. Since the other $r-1$ color graphs partition the complement of $G_i[W]$, Turán's theorem gives the full-color induced-density bound

$$e\Big(\overline{G_i[W]}\Big) \ge (r-1)p_r(|W|). \tag{4}$$

This inequality uses coexistence of all colors; it is not a consequence of the one-color local cap alone.

Choose a least color graph G. Its edge budget is

$$e(G) \le M_r := \frac{r(r^2+1)}{2}. \tag{5}$$

The shared reduction gives

$$2 \le \delta(G) \le r-1. \tag{6}$$

The upper bound uses Brooks's theorem [3]; the lower bound uses the Kang–Pikhurko extremal theorem [6].

Proof of (6). The average degree of G is at most r. If $\delta(G) = r$, then G is r-regular. On the other hand,

$$\chi(G) \ge \left\lceil \frac{r^2+1}{\alpha(G)} \right\rceil \ge r+1.$$

Some connected component therefore has chromatic number at least $r+1$. Brooks's theorem makes that component K_{r+1}, contrary to (2).

For the lower bound, suppose that a vertex v has degree $d \in \{0,1\}$ and let U be its nonneighbors other than v. Then

$$|U| = r^2 - d, \qquad \alpha(G[U]) \le r-1.$$

The complement of $G[U]$ is K_r-free. It is not $(r-1)$-partite: such a partition would have a part of order at least $r+1$, which would be a target-color K_{r+1} in $G[U]$. The Kang–Pikhurko theorem, with parameter $r-1$, gives

$$e(G[U]) \ge p_{r-1}(|U|) + \left\lfloor \frac{|U|}{r-1} \right\rfloor - 1.$$

For $d = 0$ the right side is $M_r + r + 1$. For $d = 1$ it is M_r before the edge incident with v is added. Both cases contradict (5). □

We shall also use the strict colored-density form. If $2 \le s < r$, $|W| = sr+t$, $t \ge 1$, and $\alpha(G_i[W]) \le s$, then

$$e\Big(\overline{G_i[W]}\Big) \ge (r-1)\left(p_r(|W|) + \left\lfloor \frac{|W|}{r} \right\rfloor - 1\right). \tag{7}$$

Indeed, $G_i[W]$ is not r-colorable because $|W| > rs$. For every $j \neq i$, it is a subgraph of $\overline{G_j[W]}$, so that complement is not r-partite. It is K_{r+1}-free because $\alpha(G_j) \le r$. Applying the Kang–Pikhurko theorem to every $j \neq i$ and summing proves (7).

2.1. **Colored terminal core and exact ladder.**

Theorem 2.1 (Colored terminal core)**.** *Let $r \ge 6$. There is no vertex set W of order at least $2r$ such that*

$$\alpha(G_i[W]) \le 2, \qquad \omega(G_i[W]) \le r-1.$$

Proof. Put $H = G_i[W]$, $L = \overline{H}$, and $n = |W|$. The graph L is triangle-free and $\alpha(L) \le r-1$. Every neighborhood in L is independent, whence

$$\Delta(L) \le r-1, \qquad e(L) \le \frac{n(r-1)}{2}.$$

The colored-density inequality gives

$$e(L) \ge (r-1)p_r(n).$$

At $n = 2r$ the two bounds force L to be $(r-1)$-regular. Since $r-1 > 2(2r)/5$ for $r \ge 6$, the Andrásfai–Erdős–Sós theorem [1] makes L bipartite. One side has order at least r, contrary to $\alpha(L) \le r-1$.

For $n > 2r$, one has $p_r(n) > n/2$. This follows from $p_r(2r) = r$ and

$$p_r(m+1) - p_r(m) = \left\lfloor \frac{m}{r} \right\rfloor \ge 2 \qquad (m \ge 2r).$$

The two edge bounds then contradict one another. $\square$

For fixed $r \ge 6$, define $T_2 = 0$. Once T_{s-1} is defined, put

$$A_s = (s+1)r - 2s - T_{s-1} + 1, \tag{8}$$

$$\widehat{B}_s = rs(s + T_{s-1} - 2) - 2(r-1)(s-1), \tag{9}$$

and, when $A_s > 0$ and the resulting integer is less than r, put

$$T_s = \max\left\{1, \left\lfloor \frac{\widehat{B}_s}{A_s} \right\rfloor + 1\right\}. \tag{10}$$

Theorem 2.2 (Exact colored core ladder)**.** *Whenever $T_s(r)$ is defined, there is no vertex set W with*

$$|W| \ge sr + T_s, \qquad \alpha(G_i[W]) \le s, \qquad \omega(G_i[W]) \le r-1.$$

Proof. The case $s = 2$ is Theorem 2.1. Assume the assertion at $s-1$, and put $H = G_i[W]$ and $L = \overline{H}$. The graph L is K_{s+1}-free and has independence number at most $r-1$. For each vertex x, its neighborhood in L is K_s-free and retains that independence bound. The induction hypothesis gives

$$\Delta(L) \le (s-1)r + T_{s-1} - 1 =: \Delta_s.$$

For $n = sr + t$, $1 \le t < r$, equation (7) gives

$$2e(L) \ge 2(r-1)\left(r\binom{s}{2} + st + s - 1\right).$$

After subtracting the upper bound $n\Delta_s$, the result is $A_s t - \widehat{B}_s$. Definition (10) is the first eligible integer t for which this is positive. For larger orders, the increment of the lower bound minus that of $n\Delta_s$ is at least

$$2(r-1)\left\lfloor \frac{n}{r} \right\rfloor - \Delta_s \ge A_s > 0.$$

The contradiction therefore persists for every $n \ge sr + T_s$. □

In particular,

$$T_3(r) = 1 \ (r \ge 6), \qquad T_4(r) = 2 \ (r \ge 6), \qquad (T_5(7), T_5(8)) = (5, 4). \tag{11}$$

The same argument gives the pointwise form

$$\delta(G_i[W]) \ge r + t - T_{s-1}(r) \tag{12}$$

whenever $|W| = sr + t$ and the independence and clique hypotheses at level s hold: the nonneighbors of a vertex have order at most $(s-1)r + T_{s-1} - 1$.

Let v have color-i degree $d \le r - 1$, and let U be its nonneighbor set. The colored core ladder proves the following block exclusion.

Proposition 2.3 (Shared block exclusion)**.** *The graph $G_i[U]$ cannot contain $r - 4$ pairwise disjoint copies of K_r in color i.*

Proof. Since v is anticomplete to U in color i,

$$\alpha(G_i[U]) \le r - 1. \tag{13}$$

Extend the proposed blocks to a maximal packing of k target-color copies of K_r in U, let R be the remainder, and put $s = r - k - 1$. Then

$$\alpha(G_i[R]) \le s. \tag{14}$$

For if R contained an independent $(s+1)$-set, it could be extended through the k blocks by choosing one independent representative from each. Every previously chosen vertex forbids at most one vertex in the next K_r, by the local $(r+1)$-set cap, and $(s+1) + k = r$. This would give an independent r-set in U, contrary to (13). Maximality of the packing gives

$$\omega(G_i[R]) \le r - 1. \tag{15}$$

Write $t = r - d \ge 1$. Since $|U| = r^2 - d = r(r-1) + t$, the remainder has order

$$|R| = sr + t. \tag{16}$$

If $k = r - 4$, then $s = 3$, and (14)–(16) contradict Theorem 2.2 because $T_3(r) = 1$. If $k = r-3$, the terminal-core theorem gives the same contradiction at $s = 2$. If $k = r - 2$, then (14) makes R a clique, while $|R| = r + t \ge r + 1$, a contradiction. Finally, a packing with $k \ge r - 1$ contains $r - 1$ blocks and leaves at least one vertex outside them. That vertex and one independent representative from each block give an independent r-set in U. These cases exhaust every maximal extension of the proposed $r - 4$ blocks. □

2.2. **Recursive colored edge floors.** For integers $a, m \geq 0$, let $\mathcal{F}_r(a,m)$ be the family of actual induced target-color graphs $G_i[W]$ with $|W| = m$ and

$$\alpha(G_i[W]) \leq a, \qquad \omega(G_i[W]) \leq r-1.$$

Every member and each of its induced subgraphs satisfies

$$e(G_i[Z]) \leq D_r(|Z|) := \binom{|Z|}{2} - (r-1)p_r(|Z|).$$

The full-color provenance in this definition is necessary; an abstract one-color graph satisfying the local cap need not satisfy the terminal exclusions above.

We define a lower bound $B_r(a,m)$, taking the value $+\infty$ when the recursion proves that $\mathcal{F}_r(a,m)$ is empty. Put

$$B_r(0,0) = 0, \qquad B_r(0,m) = +\infty \quad (m > 0),$$

and

$$B_r(1,m) = \begin{cases} \binom{m}{2}, & m \leq r-1, \\ +\infty, & m \geq r. \end{cases}$$

For $a \geq 2$, define

$$Q_r(a,m) = p_a(m) + \begin{cases} 0, & m \leq a(r-1), \\ \lfloor m/a \rfloor - 1, & a(r-1) < m \leq ar, \\ \lfloor m/a \rfloor, & m \geq ar+1, \end{cases} \tag{17}$$

and

$$C_r(\delta) = \begin{cases} \binom{\delta+1}{2}, & 0 \leq \delta < r-1, \\ \binom{r}{2} + 1, & \delta = r-1, \\ \delta + \left\lceil \dfrac{r+2}{r} \dbinom{\delta}{2} \right\rceil, & \delta \geq r. \end{cases} \tag{18}$$

If $T_a(r)$ is defined and $m \geq ar + T_a(r)$, set $B_r(a,m) = +\infty$. Otherwise put

$$R_r(a,m) = \min_{\substack{0 \leq \delta < m \\ B_r(a-1,m-1-\delta) < +\infty}} \max\left\{ B_r(a-1, m-1-\delta) + C_r(\delta), \left\lceil \frac{m\delta}{2} \right\rceil \right\}, \tag{19}$$

where the minimum of an empty set is $+\infty$. Put

$$b_r(a,m) = \max\{Q_r(a,m), R_r(a,m)\}.$$

Set $B_r(a,m) = +\infty$ if $b_r(a,m) > D_r(m)$; otherwise set $B_r(a,m) = b_r(a,m)$.

Proposition 2.4 (Recursive colored lower bound)**.** *Every $F \in \mathcal{F}_r(a,m)$ satisfies*

$$e(F) \geq B_r(a,m).$$

In particular, $B_r(a,m) = +\infty$ certifies that the family is empty.

Proof. Induct on a. The cases $a = 0, 1$ are immediate. For $a \geq 2$, Turán's theorem gives $e(F) \geq p_a(m)$. If $m > a(r-1)$, the complement of F is K_{a+1}-free and is not a-partite: an a-partition would give a clique of order at least r in F. Kang–Pikhurko therefore gives the middle line of (17).

When $m \geq ar+1$, equality in that bound is incompatible with the local cap. If $m \geq ar+2$, the old parts in the equality construction have total order $m-1 > ar$, so one part gives a clique of order at least $r+1$. If $m = ar+1$, all old parts have order r. In the notation of the equality construction, a special old part S, exceptional vertex x, witness vertex y, and nonempty proper $A \subset S$ give

$$e(F[S \cup \{x\}]) = \binom{r}{2} + r - |A|, \qquad e(F[S \cup \{y\}]) = \binom{r}{2} + |A|.$$

The local cap would require both $r - |A| \leq 1$ and $|A| \leq 1$, which is impossible for $r \geq 6$. This proves $e(F) \geq Q_r(a, m)$.

Choose a minimum-degree vertex x, set $\delta = d_F(x)$, $N = N_F(x)$, and $U = V(F) \setminus (N \cup \{x\})$. Then $F[U] \in \mathcal{F}_r(a-1, m-1-\delta)$. Put

$$X = e_F(N, U), \qquad Y = e_F(N).$$

Minimum degree gives $X + 2Y \geq \delta(\delta - 1)$. As $Y \leq \binom{\delta}{2}$, this first gives $X + Y \geq \binom{\delta}{2}$. If $\delta = r - 1$, equality would force $Y = \binom{r-1}{2}$ and $X = 0$, making $N \cup \{x\}$ a K_r; one extra edge is therefore required.

If $\delta \geq r$, average the local cap over the sets $\{x\} \cup S$ with $S \in \binom{N}{r}$. Each S spans at most $\binom{r-1}{2}$ edges, and hence

$$Y \leq \frac{r-2}{r}\binom{\delta}{2}.$$

Combining this with the degree-sum inequality gives

$$\delta + X + Y \geq C_r(\delta).$$

The induction hypothesis on $F[U]$ proves the first term in the maximum in (19); the degree sum gives $e(F) \geq \lceil m\delta/2 \rceil$. Taking both bounds at the same δ, then minimizing over possible minimum degrees, proves (19). Theorem 2.2 justifies each terminal $+\infty$ value, and $e(F) \leq D_r(m)$ justifies the final emptiness test. $\square$

For a least color graph G_i, let v be a minimum-degree vertex of degree $d \leq r - 1$, and let U be its nonneighbor set. After selecting j disjoint target copies of K_r in U, let R_j be their complement. If R_j has no further K_r, representative selection gives

$$R_j \in \mathcal{F}_r(r-1-j, r^2 - d - jr).$$

Indeed, an independent $(r-j)$-set in R_j could be extended by one representative from each deleted block. At every choice, fewer than r previously selected vertices each forbid at most one block vertex, so a representative remains. The resulting independent r-set in U contradicts (13).

For the edge accounting, put $N = N_{G_i}(v)$, $X = e_{G_i}(N, U)$, and $Y = e_{G_i}(N)$. Minimum degree gives $X + 2Y \geq d(d-1)$, while $Y \leq \binom{d}{2}$; hence $X + Y \geq \binom{d}{2}$. Subtracting the edges incident with v, these neighborhood edges, and the internal edges of the selected blocks gives

$$e(R_j) \leq E_{r,d,j} := M_r - d - \binom{d}{2} - j\binom{r}{2}. \tag{20}$$

Consequently,

$$B_r(r-1-j, r^2 - d - jr) > E_{r,d,j} \tag{21}$$

forces every packing of j blocks to extend by one block.

The exact margins $B_r - E_{r,d,j}$ needed below are

r	d	$j=0$	$j=1$	$j=2$
7	0	57	$+\infty$	$+\infty$
7	1	42	$+\infty$	$+\infty$
7	2	25	$+\infty$	$+\infty$
7	3	18	28	$+\infty$
7	4	10	9	$+\infty$
7	5	5	4	$+\infty$
7	6	-1	-2	-3

d	$j=0$	$j=1$	$j=2$	$j=3$
0	60	104	$+\infty$	$+\infty$
1	44	72	$+\infty$	$+\infty$
2	35	41	$+\infty$	$+\infty$
3	24	23	$+\infty$	$+\infty$
4	17	16	$+\infty$	$+\infty$
5	8	7	6	$+\infty$
6	3	2	1	$+\infty$
7	-4	-5	-6	-7

The left table is for $r = 7$ and the right table for $r = 8$. Every entry needed for $d \le 5$ at $r = 7$, and for $d \le 6$ at $r = 8$, is positive or infinite. Criterion (21) therefore forces, respectively, three or four disjoint blocks, contrary to Proposition 2.3. Together with (6), this proves

$$\delta(G) = 6 \quad (r = 7), \qquad \delta(G) = 7 \quad (r = 8). \tag{22}$$

3. The seven-color case

Theorem 3.1. *Every seven-coloring of K_{50} has an eight-vertex set that omits at least one color.*

The only new finite premise for this theorem is the next weighted seven-vertex lemma.

Lemma 3.2 (Weighted light edge)**.** *Let M be a graph on seven vertices with*

$$\alpha(M) \le 5, \qquad 6 \le m := e(M) \le 9.$$

Give its vertices nonnegative integer weights z_y, and put $Z = \sum_y z_y$.

(i) *If $m + Z \le 8$, some edge yy' satisfies*

$$d_M(y) + d_M(y') + z_y + z_{y'} \le 6.$$

(ii) *If $m + Z \le 9$, some edge yy' satisfies*

$$d_M(y) + d_M(y') + z_y + z_{y'} \le 7.$$

The direct finite check gives the exact extrema

$m\backslash Z$	0	1	2	3
6	6	6	6	7
7	6	6	7	
8	6	6		
9	7			

over all cases satisfying the hypotheses. Section 4 states the semantics and independent reconstruction.

3.1. **Human implication chain.** Assume that every eight-set in a seven-coloring of K_{50} sees all seven colors, and choose a least color graph G.

R7.1 The least-color budget is $e(G) \le 175$.

R7.2 The shared colored recursion excludes minimum degree at most five, while (6) gives the upper bound six. Hence $\delta(G) = 6$.

R7.3 For a degree-six vertex, its 43 nonneighbors U satisfy

$$\alpha(G[U]) \le 6, \qquad e(G[U]) \le 154.$$

R7.4 Lemma 3.2, the star-and-cover argument, and the shared recursion give

$$P_6(43) \ge 156, \qquad P_5(36) \ge 134, \qquad P_4(29) \ge 113.$$

R7.5 These floors exceed the successive edge upper bounds

$$154, \qquad 133, \qquad 112.$$

Three disjoint target copies of K_7 are therefore forced in U.

R7.6 Proposition 2.3 forbids exactly three such blocks when $r = 7$, giving the contradiction.

The residual ledger is

residual order	independence cap	edge upper	no-next-K_7 floor
43	6	154	156
36	5	133	134
29	4	112	113

For this section, $P_a(n)$ denotes a valid lower edge bound on an actual induced target-color graph F with $|F| = n$, $\alpha(F) \le a$, and $\omega(F) \le 6$. The recursive bound above gives

$$\begin{gathered} P_3(20) \ge 63, \qquad P_3(21) \ge 75, \\ P_5(37) \ge 143, \qquad P_5(38) \ge 153, \qquad P_5(39) \ge 176. \end{gathered} \tag{23}$$

3.2. **The star-and-cover mechanism.** Let x be a minimum-degree vertex of an induced target graph F, and put

$$\delta = d_F(x), \qquad N = N_F(x), \qquad W = V(F) \setminus N_F[x].$$

Write $M = \overline{F[N]}$ and $m = e(M)$. For $y \in N$, minimum degree gives $d_F(y, W) \ge d_M(y)$. Define nonnegative integers z_y by

$$d_F(y, W) = d_M(y) + z_y, \qquad Z = \sum_{y \in N} z_y.$$

Then

$$e_F(N, W) = 2m + Z. \tag{24}$$

If $\delta = 6$, the number of edges outside $F[W]$ is

$$6 + e_F(N) + e_F(N, W) = 21 + m + Z. \tag{25}$$

Here M is nonempty, since otherwise $N_F[x]$ is a K_7. If $\delta = 7$, the corresponding total is

$$28 + m + Z. \tag{26}$$

The local eight-set cap applied to $N_F[x]$ gives $m \ge 6$. Also $\alpha(M) \le 5$, since an independent six-set in M, together with x, would be a target K_7.

For an edge yy' of M, put

$$C = N_F(y, W) \cup N_F(y', W).$$

Then

$$|C| \le d_M(y) + d_M(y') + z_y + z_{y'}. \tag{27}$$

When $\alpha(F) \le 4$, the graph $F[W \setminus C]$ has independence number at most two: an independent triple there, together with the nonadjacent pair y, y', would be an independent five-set in F. Thus $|W \setminus C| \ge 14$ contradicts Theorem 2.1.

When $\delta = 6$, choose any edge yy' of the nonempty graph M. The quantity $d_M(y) + d_M(y') - 1$ counts the edges incident to at least one of y, y', so

$$d_M(y) + d_M(y') \le m + 1.$$

Together with $z_y + z_{y'} \le Z$, this bounds the right side of (27) by $m + 1 + Z$.

Lemma 3.3. *One has*

$$P_4(28) \ge 100, \qquad P_4(29) \ge 113.$$

Proof. Suppose first that $|F| = 28$ and $e(F) \le 99$. Average degree gives $\delta \le 7$. If $\delta \le 5$, then $|W| \ge 22 = 3 \cdot 7 + T_3(7)$, contrary to Theorem 2.2.

If $\delta = 6$, then $|W| = 21$ and (23) gives $e(F[W]) \ge 75$. Equation (25) gives $m + Z \le 3$; the bound following (27) gives $|C| \le 4$. Hence $|W \setminus C| \ge 17$, a contradiction.

If $\delta = 7$, then $|W| = 20$ and $e(F[W]) \ge 63$. Equation (26) gives $m + Z \le 8$. Lemma 3.2 gives $|C| \le 6$, so $|W \setminus C| \ge 14$, again a contradiction. This proves the first floor.

Now suppose that $|F| = 29$ and $e(F) \le 112$. Average degree gives $\delta \le 7$. A degree at most six leaves at least 22 nonneighbors and is excluded by the three-layer theorem. Thus $\delta = 7$, $|W| = 21$, and $e(F[W]) \ge 75$. Equation (26) gives $m + Z \le 9$. Lemma 3.2 gives $|C| \le 7$, leaving at least 14 vertices and producing the terminal-core contradiction. □

Lemma 3.4. *One has*

$$P_5(35) \ge 122, \qquad P_5(36) \ge 134, \qquad P_6(43) \ge 156.$$

Proof. If a 35-vertex graph with independence number at most five has at most 121 edges, its minimum degree is at most six. A degree at most four leaves at least 30 nonneighbors and is excluded by $T_4(7) = 2$. At degrees five and six, Lemma 3.3 and the star floors give

$$P_4(29) + 15 \ge 128, \qquad P_4(28) + 22 \ge 122.$$

This proves $P_5(35) \ge 122$.

For order 36 and at most 133 edges, the minimum degree is at most seven. Degrees at most five are excluded by the four-layer theorem. At degrees six and seven, the lower bounds are

$$P_4(29) + 22 \ge 135, \qquad P_4(28) + 34 \ge 134.$$

This proves $P_5(36) \ge 134$.

Finally, suppose that $|F| = 43$, $\alpha(F) \le 6$, and $e(F) \le 155$. The minimum degree is at most seven. A degree at most two leaves at least 40 nonneighbors and is excluded by $T_5(7) = 5$. For degrees three through seven the lower bounds are

δ	3	4	5	6	7
$e(F)$	$176 + 6$	$153 + 10$	$143 + 15$	$134 + 22$	$122 + 34.$

Every entry is at least 156, proving the final floor. □

Proof of Theorem 3.1. Suppose otherwise and choose a least color graph G. Equation (22) gives $\delta(G) = 6$. Fix a degree-six vertex v, and let U be its 43 nonneighbors. Then $\alpha(G[U]) \leq 6$. Minimum-degree accounting gives at least $6 + \binom{6}{2} = 21$ target edges outside U, and hence

$$e(G[U]) \leq 175 - 6 - \binom{6}{2} = 154.$$

If $G[U]$ had no K_7, Lemma 3.4 would give at least 156 edges. Choose a target K_7, delete it, and call the 36-vertex remainder R_1. If R_1 had an independent six-set, each of its vertices would forbid at most one vertex of the deleted K_7, so one block vertex could be adjoined to form an independent seven-set in U. Hence $\alpha(G[R_1]) \leq 5$, while

$$e(G[R_1]) \leq 154 - \binom{7}{2} = 133.$$

Lemma 3.4 forces a second disjoint K_7. Delete it and call the 29-vertex remainder R_2. An independent five-set in R_2 could be extended first through one deleted block and then through the other, with at most five and then six forbidden block vertices. Thus

$$\alpha(G[R_2]) \leq 4, \qquad e(G[R_2]) \leq 133 - \binom{7}{2} = 112.$$

Lemma 3.3 forces a third disjoint K_7. Thus U contains three disjoint target copies of K_7, contrary to Proposition 2.3. $\square$

4. Finite verification for the seven-color case

A finite case is a pair (M, z), where M is a labeled graph on seven vertices, $6 \leq e(M) \leq 9$, $\alpha(M) \leq 5$, and $z \in \mathbb{N}^7$ satisfies $e(M) + \sum_y z_y \leq 9$. For each case, the checker computes

$$\min_{yy' \in E(M)} \left(d_M(y) + d_M(y') + z_y + z_{y'}\right).$$

The exact enumeration is as follows.

(m, Z)	weighted cases	largest possible minimum
$(6,0)$	54,257	6
$(6,1)$	379,799	6
$(6,2)$	1,519,196	6
$(6,3)$	4,557,588	7
$(7,0)$	116,280	6
$(7,1)$	813,960	6
$(7,2)$	3,255,840	7
$(8,0)$	203,490	6
$(8,1)$	1,424,430	6
$(9,0)$	293,930	7
total	12,618,770	

The admissible labeled graph counts for $m = 6, 7, 8, 9$ are respectively

$$54{,}257, \quad 116{,}280, \quad 203{,}490, \quad 293{,}930.$$

The canonical checker enumerates labeled edge subsets and weak compositions directly. The separate Sage checker uses nauty [7] to generate 40, 65, 97, and 131 unlabelled graph types, then reconstructs the labeled counts from automorphism orders. Both routes reproduce the table. Neither route calls a SAT solver, and there is no detached proof trace.

The accompanying release contains the direct checker `code/verify_r7_weighted_light_edge.py`, the independent Sage checker `code/verify_r7_weighted_light_edge_sage.py`, the arithmetic replay `code/verify_r7_fixed_outer_closure.py`, and the shared ladder check `code/verify_colored_core_ladder.py`. Their hashes are recorded in the release checksum ledger.

5. The eight-color case

Theorem 5.1. *Every eight-coloring of K_{65} has a nine-vertex set that omits at least one color.*

Throughout this section, $P_a(n)$ denotes a valid lower edge bound on an actual induced target-color graph with n vertices, independence number at most a, and clique number at most seven. The proof needs two such edge floors.

Lemma 5.2 (Twenty-three-vertex full-color floor)**.** *Let H be an actual induced target-color graph in a hypothetical balanced eight-coloring. If*

$$|V(H)| = 23, \qquad \alpha(H) \le 3, \qquad \omega(H) \le 7,$$

then $e(H) \ge 91$.

The proof reduces a putative graph with at most 90 edges to a degree-seven vertex with a 15-vertex triangle-free complement core. Full-color density and the Kang–Pikhurko theorem force the extremal core. The missing-edge graph on the seven-vertex neighborhood is a star, and each center–leaf row pair partitions a minimum cover of the core. The full-color ten-set cap removes one split core; the local nine-set cap removes the remaining two.

Proof. Suppose instead that $e(H) \le 90$, and choose a minimum-degree vertex v. Average degree gives $d_H(v) \le \lfloor 180/23 \rfloor = 7$. If $d_H(v) \le 6$, its nonneighbors contain at least 16 vertices. Their induced graph has independence number at most two and clique number at most seven, contrary to Theorem 2.1. Hence $d_H(v) = 7$.

Put $A = N_H(v)$ and $B = V(H) \setminus (A \cup \{v\})$, so $|A| = 7$ and $|B| = 15$. Define

$$L = \overline{H[B]}, \qquad M = \overline{H[A]}, \qquad m = e(L), \qquad f = e(M).$$

The graph L is triangle-free, has $\alpha(L) \le 7$, and is nonbipartite. Indeed, one side of a bipartition of 15 vertices would have order at least eight and would give a target K_8 in $H[B]$. The colored-density bound gives

$$m \ge 7p_8(15) = 49. \tag{28}$$

The Kang–Pikhurko bound for a nonbipartite triangle-free graph on 15 vertices gives

$$m \le t_2(15) - \left\lfloor \frac{15}{2} \right\rfloor + 1 = 56 - 7 + 1 = 50. \tag{29}$$

Here $t_2(15) = \lfloor 15^2/4 \rfloor = 56$ is the triangle-free Turán number.

For $a \in A$, put

$$D_a = N_H(a) \cap B, \qquad c_a = |D_a|, \qquad c = \sum_{a \in A} c_a.$$

Minimum degree gives

$$d_H(a) = 1 + 6 - d_M(a) + c_a \geq 7, \qquad c_a \geq d_M(a), \qquad c \geq 2f.$$

Also $f \geq 1$, since otherwise $\{v\} \cup A$ is a target K_8. The full edge ledger is

$$e(H) = 7 + (21 - f) + (105 - m) + c = 133 - f - m + c. \tag{30}$$

It follows that

$$c \leq m + f - 43 \leq f + 7, \qquad f \leq 7. \tag{31}$$

For every edge $aa' \in E(M)$, the set $D_a \cup D_{a'}$ is a vertex cover of L. Otherwise an uncovered edge of L, together with a, a', would be an independent four-set in H. Since $\tau(L) = 15 - \alpha(L) \geq 8$, we have

$$c_a + c_{a'} \geq 8 \qquad (aa' \in E(M)). \tag{32}$$

The graph M has no two disjoint edges, because they would give $c \geq 16$, while (31) gives $c \leq 14$. If M contained a triangle, the absence of two disjoint edges would make that triangle its entire nonisolated part, so $f = 3$. Summing (32) around the triangle would give $c \geq 12$, while (31) would give $c \leq 10$.

A nonempty graph with matching number at most one and no triangle is a star. Thus

$$M = K_{1,f} \cup (6 - f)K_1, \qquad 1 \leq f \leq 6.$$

Let a_0 be the center, $a_1, \ldots, a_f$ the leaves, and put $x = c_{a_0}$. Minimum degree and (32) give

$$x \geq f, \qquad c_{a_j} \geq \max\{1, 8 - x\} \qquad (1 \leq j \leq f).$$

Consequently,

$$c \geq x + f \max\{1, 8 - x\} \geq f + 7. \tag{33}$$

For $x \leq 7$, the difference between the final two quantities is $(f - 1)(7 - x)$; for $x \geq 7$, it is $x - 7$. Equations (29), (31), and (33) force

$$m = 50, \qquad c = f + 7, \qquad e(H) = 90. \tag{34}$$

For every leaf a_j, equality also gives

$$c_{a_0} + c_{a_j} = 8, \qquad D_{a_0} \cap D_{a_j} = \varnothing,$$

and $C_j := D_{a_0} \cup D_{a_j}$ is an eight-vertex minimum cover of L. For $f = 1$ this follows from $c = 8$ and (32). For $f \geq 2$, equality in (33) forces $x = 7$, every leaf row to have size one, and every isolated row to be empty.

The equality case of the Kang–Pikhurko theorem now applies. There are disjoint sets P, Q and vertices x_0, y_0 such that

$$|P| = 6, \quad |Q| = 7, \quad Q = X \mathbin{\dot\cup} Y, \quad a := |X| \in \{1, 2, 3\},$$

and

$$E(L) = E(K_{P,Q}) \cup \{x_0 y_0\} \cup \{x_0 q : q \in X\} \cup \{y_0 q : q \in Y\}.$$

Swapping x_0, y_0 permits $a \leq 3$. The maximum independent seven-sets and their complementary minimum covers are exactly

independent set	minimum cover C	$e_L(C)$
Q	$P \cup \{x_0, y_0\}$	1
$P \cup \{x_0\}$	$Q \cup \{y_0\}$	$7 - a$
$P \cup \{y_0\}$	$Q \cup \{x_0\}$	a
$\{x_0\} \cup Y$	$P \cup X \cup \{y_0\}$	6, if $a = 1$

This list is exhaustive. An independent set meeting P cannot meet Q, and order seven then forces all six vertices of P and one of x_0, y_0. An independent set avoiding P is checked directly from $Q = X \dot{\cup} Y$.

Fix a leaf a_j. The ten-set $S = \{a_0, a_j\} \cup C_j$ contains no target edge between a_0, a_j, exactly eight cross edges, and $28 - e_L(C_j)$ target edges inside C_j. The full-color ten-set density bound, not merely the local nine-set cap, therefore gives

$$36 - e_L(C_j) = e_H(S) \le D_8(10) = 31, \qquad e_L(C_j) \ge 5. \tag{35}$$

If $a = 3$, every minimum cover in the table spans at most four edges, contrary to (35). If $a = 2$, the only remaining cover is $C_j = Q \cup \{y_0\}$. Put $C' = Q \cup \{x_0\}$; then $e_L(C') = 2$ and $|C_j \cap C'| = 7$. For each $z \in \{a_0, a_j\}$, the local cap on the nine-set $\{z\} \cup C'$ gives

$$28 - 2 + |D_z \cap C'| \le 29, \qquad |D_z \cap C'| \le 3.$$

The two rows partition C_j, so their intersections with C' sum to seven, not at most six.

Finally, let $a = 1$. If $C_j = Q \cup \{y_0\}$, use $C' = Q \cup \{x_0\}$. If $C_j = P \cup X \cup \{y_0\}$, use $C' = P \cup \{x_0, y_0\}$. In both cases

$$e_L(C') = 1, \qquad |C_j \cap C'| = 7.$$

The local nine-set cap gives $|D_z \cap C'| \le 2$ for each of the two rows, again contradicting their total intersection of seven. All equality cores are impossible. □

The second floor is the computer-assisted dependency

$$P_3(24) \ge 106. \tag{36}$$

Its exhaustive minimum-degree split is

minimum degree	reduction	receipt
$0, \ldots, 4$	complement degree count	human arithmetic
5	one 18-vertex endpoint formula	semantic reconstruction and one LRAT
6	17-vertex complement degree argument	human proof
7	16-vertex complement–shell incidence argument	human proof
8	861 fixed core–shell pairs	semantic reconstruction and 861 LRATs

Proof of (36). Suppose that H is a counterexample: $|V(H)| = 24$, $\alpha(H) \le 3$, $\omega(H) \le 7$, every nine-set spans at most 29 edges, and $e(H) \le 105$. Choose a minimum-degree vertex v of degree $d \le 8$, let U be its $u = 23 - d$

nonneighbors, and put $L = \overline{H[U]}$. The graph L is triangle-free, $\alpha(L) \le 7$, every nine-set of L spans at least seven edges, and $\Delta(L) \le 7$.

Minimum-degree accounting gives

$$e(H) \ge \binom{d+1}{2} + \binom{u}{2} - e(L). \tag{37}$$

Using $e(L) \le \lfloor 7u/2 \rfloor$ gives

d	u	lower bound for $e(H)$	conclusion
0	23	173	impossible
1	22	155	impossible
2	21	140	impossible
3	20	126	impossible
4	19	115	impossible
5	18	105	equality endpoint

At $d = 5$, equality in (37) forces $e(L) = 63$, so L is seven-regular. Equality in the star accounting forces $H[N_H[v]]$ to be a K_6 with no edges from $N_H(v)$ to U. Thus the branch requires a triangle-free seven-regular graph on 18 vertices with independence number at most seven. The endpoint formula in Section 6 encodes exactly that graph and has a checked LRAT refutation.

If $d = 6$, then $u = 17$. In fact $\Delta(L) \le 6$. To see this, suppose that z has seven neighbors A in L; A is independent. Among the remaining nine vertices b, put $p(b) = |N_L(b) \cap A|$. For distinct b, c, the nine-set $A \cup \{b, c\}$ gives

$$p(b) + p(c) + \mathbf{1}_{bc \in E(L)} \ge 7.$$

At most two vertices have $p(b) \le 3$: two such vertices must both have value three and be adjacent, while three would form a triangle. Any two vertices with p-value at least four are nonadjacent, because their neighborhoods in the seven-set A intersect. At least seven such vertices, together with z, form an independent eight-set, a contradiction. Hence $e(L) \le 51$, and

$$e(H) \ge \binom{7}{2} + \binom{17}{2} - 51 = 106.$$

It remains to exclude $d = 7, 8$.

The degree-seven branch. Here $|A| = 7$ and $|B| = 16$ for $A = N_H(v)$ and $B = V(H) \setminus (A \cup \{v\})$. Put $L = \overline{H[B]}$. It is triangle-free, has $\alpha(L) \le 7$, every nine vertices span at least seven edges, and $\Delta(L) \le 7$.

We first prove $e(L) \le 48$. It is enough to show that a degree-seven vertex forces $e(L) = 44$. Suppose z has degree seven, put $P = N_L(z)$, and let Q be the other eight vertices. The set P is independent. For $b \in Q$, write $p_b = |N_L(b) \cap P|$. For distinct $b, c \in Q$,

$$p_b + p_c + \mathbf{1}_{bc \in E(L)} \ge 7. \tag{38}$$

At most two vertices are low, meaning $p_b \le 3$. Any two low vertices must have value three and be adjacent, while three low vertices would form a triangle. Any two high vertices are nonadjacent, since their neighborhoods in P both have order at least four. There cannot be seven high vertices, since those vertices together with z would be an independent eight-set. Thus Q consists of six high vertices and two low vertices x, y, with

$$p_x = p_y = 3, \qquad xy \in E(L).$$

The only possible edges inside Q are xy and edges from x or y to the six high vertices. No high vertex can meet both x and y, so $e(L[Q]) \leq 7$. The nine-set $\{z\} \cup Q$ gives the reverse inequality, and every high vertex meets exactly one of x, y. Let R be the six high vertices, and put $R_x = N_L(x) \cap R$ and $R_y = N_L(y) \cap R$. Then R_x, R_y partition R, and the degree bound gives $|R_x| = |R_y| = 3$. For $r \in R_x$, the sets $N_L(r) \cap P$ and $N_L(x) \cap P$ are disjoint; their orders are at least four and exactly three, so they partition P. The same holds on the y side. Consequently

$$\sum_{b \in Q} p_b = 2 \cdot 3 + 6 \cdot 4 = 30, \qquad e(L) = 7 + 30 + 7 = 44.$$

If no degree-seven vertex exists, $e(L) \leq 16 \cdot 6/2 = 48$.

Put $F = \overline{H[A]}$, $f = e(F)$, and, for $a \in A$, put

$$D_a = N_H(a) \cap B, \qquad c_a = |D_a|, \qquad c = \sum_{a \in A} c_a.$$

The graph F is nonempty, or $H[\{v\} \cup A]$ is a K_8. Minimum degree gives $c_a \geq d_F(a)$ and $c \geq 2f$. The full edge count is

$$e(H) = 148 - m - f + c \leq 105, \qquad m = e(L),$$

so

$$c \leq m + f - 43 \leq f + 5. \tag{39}$$

For every $aa' \in E(F)$, the set $D_a \cup D_{a'}$ covers L; an uncovered L-edge would make an independent four-set with a, a'. Since $\tau(L) \geq 9$,

$$c_a + c_{a'} \geq 9 \qquad (aa' \in E(F)). \tag{40}$$

Equations (39)–(40) give $f \geq 4$, while $c \geq 2f$ and (39) give $f \leq 5$.

The graph F cannot have two disjoint edges, since (40) would give $c \geq 18$, whereas (39) gives $c \leq 10$. A graph with matching number at most one is a star or a triangle together with isolated vertices. Since $f \in \{4, 5\}$, it is a star $K_{1,f}$. If a_0 is its center and $x = c_{a_0}$, then

$$c \geq x + f \max\{1, 9 - x\} \geq f + 8,$$

contrary to (39). This excludes $d = 7$.

The degree-eight branch. Now $|A| = 8$ and $|B| = 15$. Define

$$L = \overline{H[B]}, \quad F = \overline{H[A]}, \quad m = e(L), \quad f = e(F).$$

The graph L is triangle-free, $\alpha(L) \leq 7$, $\Delta(L) \leq 7$, and every nine-set spans at least seven edges. The graph F is K_4-free and $\alpha(F) \leq 6$. Applying the local cap to $\{v\} \cup A$ gives $f \geq 7$.

For $a \in A$, write

$$D_a = N_H(a) \cap B, \qquad c_a = d_F(a) + \varepsilon_a, \qquad \varepsilon_a \geq 0.$$

The equality follows from minimum degree. With $c = \sum_a c_a = 2f + \sum_a \varepsilon_a$, the edge ledger is

$$e(H) = 8 + (28 - f) + (105 - m) + c = 141 - f - m + c \leq 105.$$

Hence

$$c \le m + f - 36, \qquad \sum_a \varepsilon_a \le m - f - 36. \tag{41}$$

The first inequality and $f \ge 7$ give $m \ge 43$. The graph L is nonbipartite, because a bipartition has a part of order at least eight. Kang–Pikhurko gives $m \le 50$, so

$$43 \le m \le 50. \tag{42}$$

Two incidence conditions are forced. If $aa' \in E(F)$, the rows $D_a, D_{a'}$ cover L, and therefore

$$d_F(a) + d_F(a') + \varepsilon_a + \varepsilon_{a'} \ge 8. \tag{43}$$

If T is a triangle of F, the three corresponding rows cover all of B, and therefore

$$\sum_{a \in T} (d_F(a) + \varepsilon_a) \ge 15. \tag{44}$$

Nauty 2.8.9's `geng` stream [7] contains all unlabelled triangle-free graphs on 15 vertices with maximum degree at most seven and 43 through 50 edges. There are 46,184 stream records. Exact checks of $\alpha(L) \le 7$ and the nine-set lower bound retain 997 core types, distributed as

m	43	44	45	46	47	48	49	50
cores	435	250	152	82	45	21	9	3.

Equation (41) also gives $f \le m - 36 \le 14$. The shell stream therefore contains all 6,896 unlabelled eight-vertex graphs with 7 through 14 edges. It rejects K_4 or an independent seven-set and, for each remaining F, enumerates every weak composition of $\sum_a \varepsilon_a \le 7$. The bound seven follows from $f \ge 7$, $m \le 50$, and (41). It then checks (41), (43), and (44). Thus every possible fixed pair from this branch occurs in the two filtered streams.

If the triangle condition is temporarily omitted, the other filters leave no shell through $m = 46$. At $m = 47$ they leave three book graphs: two adjacent centers, $t = 3, 4, 5$ common leaves, and $5 - t$ isolated vertices. The forced row-size vector is

$$6, 6, \underbrace{2, \ldots, 2}_{t \text{ times}}, \underbrace{0, \ldots, 0}_{5-t \text{ times}}.$$

A center–center–leaf triangle has row-size sum 14, contrary to (44). Hence no case with $m \le 47$ remains. For $m = 48, 49, 50$, respectively, the shell filter retains 17, 40, and 48 records, while the core filter retains 21, 9, and 3 types. It follows that the finite index contains exactly

$$21 \cdot 17 + 9 \cdot 40 + 3 \cdot 48 = 861$$

core–shell pairs.

For a fixed pair, the 120 Boolean variables record membership $b \in D_a$. The formula enforces the global edge cap; the row bounds $|D_a| \ge d_F(a)$; the column bounds $|\{a : b \in D_a\}| \ge \max\{0, d_L(b) - 6\}$; the mixed independent-four exclusions; the triangle-column covers; and the mixed K_8 exclusions. It also enforces the local-cap families consisting of one shell and eight core vertices, of v with six shell and two core vertices, and of v with seven shell and one core vertex. Every graph in the branch satisfies these constraints, so unsatisfiability of this restricted formula is enough; the encoding need not include every other nine-set family. The standard-library semantic verifier independently reconstructs the raw filters, the pair order, every clause family, and every auxiliary totalizer. All 861 reconstructed formulas are refuted by checked LRAT proofs, as detailed in Section 6. Thus $d = 8$ is also impossible, completing the proof of (36). □

5.1. **Human implication chain.** Assume that every nine-set in an eight-coloring of K_{65} sees all eight colors, and choose a least color graph G.

R8.1 The least-color budget is $e(G) \leq 260$.

R8.2 The shared colored recursion excludes minimum degrees at most six, and (6) gives $\delta(G) = 7$.

R8.3 For a degree-seven vertex, its 57 nonneighbors U satisfy

$$\alpha(G[U]) \leq 7, \qquad e(G[U]) \leq 232.$$

R8.4 Lemma 5.2, (36), and the shared recursion give

$$P_7(57) \geq 235, \quad P_6(49) \geq 206, \quad P_5(41) \geq 177, \quad P_4(33) \geq 149.$$

R8.5 These values exceed the successive edge upper bounds

$$232, \quad 204, \quad 176, \quad 148.$$

Four disjoint target copies of K_8 are forced in U.

R8.6 Proposition 2.3 forbids exactly four such blocks when $r = 8$.

The full residual table is

j	order	independence cap	edge upper	no-next-K_8 floor
0	57	7	232	235
1	49	6	204	206
2	41	5	176	177
3	33	4	148	149

For actual induced target graphs with clique number at most seven, importing Lemma 5.2 and (36) into Proposition 2.4 gives

a	orders n	corresponding floors $P_a(n)$
4	31, 32, 33	120, 134, 149
5	40, 41, 42, 43	163, 177, 189, 203
6	48, 49, 50	192, 206, 218
7	57	235

The verifier `verify_r8_d7_full_color_bridge.py` evaluates the recursion and records every minimizing degree branch.

Proof of Theorem 5.1. Suppose otherwise and choose a least color graph G. Equation (22) gives $\delta(G) = 7$. Choose a degree-seven vertex v and let U be its nonneighbors. Minimum-degree accounting gives

$$|U| = 57, \qquad \alpha(G[U]) \leq 7, \qquad e(G[U]) \leq 260 - 7 - \binom{7}{2} = 232.$$

After deleting j disjoint target copies of K_8, let R_j be the residual. Then

$$|R_j| = 57 - 8j, \qquad \alpha(G[R_j]) \leq 7 - j, \qquad e(G[R_j]) \leq 232 - 28j. \tag{45}$$

The independence claim follows by representative selection. If R_j had an independent set of order $8 - j$, choose one representative from each deleted block in sequence. Every outside vertex has at most one target neighbor in a target K_8, since two such neighbors would violate the local nine-set cap. Before each choice at most seven vertices have been selected, so at most seven vertices of the next block are forbidden. The result is an independent eight-set in U; together with v it contradicts $\alpha(G) \leq 8$.

Every R_j is obtained only by vertex deletion, so it remains an actual induced subgraph of the original color graph. In particular, it retains the full-color provenance required by Lemma 5.2 and Proposition 2.4. For $j = 0, 1, 2, 3$, the four floors in the preceding table are strictly above the respective upper bounds $232, 204, 176, 148$. Therefore R_j contains a target K_8 at each stage. Four disjoint target copies of K_8 are forced in U, contrary to Proposition 2.3. $\square$

6. Finite verification for the eight-color case

For the degree-five endpoint, the formula searches for a labeled graph on 18 vertices that is 7-regular, triangle-free, and has independence number at most seven, after fixing one neighborhood by symmetry. For degree eight, write

$$A = N_H(v), \quad |A| = 8, \qquad B = V(H) \setminus (A \cup \{v\}), \quad |B| = 15,$$

and put $L = \overline{H[B]}$ and $F = \overline{H[A]}$. The human reduction filters the unlabelled core and shell streams, then indexes every pair satisfying the remaining edge ledger.

finite branch	instances	full clauses	LRATs
degree five	1	50,225	1
degree eight	861	52,434,647	861
total	862	52,484,872	862

The degree-five formula has 1,233 variables and 288 totalizer merges. Across the degree-eight formulas, the semantic verifier reconstructs 413,938 retained clauses and 836,035 totalizer merges.

The degree-eight stream reduction is

$e(L)$	filtered core types	filtered shell records	indexed pairs
48	21	17	357
49	9	40	360
50	3	48	144
total	33	105	861

There are no indexed pairs with $e(L) \leq 47$. Before the final pairing, nauty 2.8.9 produces 46,184 raw 15-vertex core graphs and 6,896 raw eight-vertex shell graphs. The semantic filters retain 997 core types and 905 shell records across all edge rows. The two exact generator commands are

```
geng -q -t -D7 15 43:50
geng -q 8 7:14
```

The local `geng` binary used to create the retained streams has SHA-256

`3ca950af2145c546f9f586cf960eaf98`
`f88fc3920564338f8306b6f58d018af5`.

The accompanying release contains the degree-five semantic verifier `code/verify_endpoint18_certificate.py`, its compressed CNF and LRAT in `artifacts/endpoint18/`, and the complete degree-eight package under `artifacts/r8-fixed-outer/`.

The degree-eight portable archive has 2,600 regular files and a 2,599-entry checksum ledger. Its size is 15,226,513 bytes and its SHA-256 is

`af5e7d7c20ab44f2330e7b69fdffe16c`
`018eb948bfb41539ff2f0a7f2e14bfc0`.

7. Replay and trust boundary

From the `r7-r8/` release directory, the seven-color replay surface is

```
python3 code/verify_r7_weighted_light_edge.py
sage code/verify_r7_weighted_light_edge_sage.py
python3 code/verify_r7_fixed_outer_closure.py
python3 code/verify_colored_core_ladder.py
```

The pinned LRAT checker and the full eight-color replay can be built and run with

```
git clone https://github.com/marijnheule/drat-trim.git
git -C drat-trim checkout 2e3b2dc0ecf938addbd779d42877b6ed69d9a985
make -C drat-trim lrat-check
./reproduce.sh --lrat-check "$PWD/drat-trim/lrat-check"
```

The LRAT format and its checking semantics are described in [4]; the DRAT-trim toolchain is described in [8]. The audited `lrat-check` executable was built from DRAT-trim source commit

`2e3b2dc0ecf938addbd779d42877b6ed69d9a985`.

Its arm64 binary SHA-256 is

`a2caddba197bbb0846aabf2dd54e87d7`
`9612e3b840ea0c43714f5c522f2d86c7`.

The finite checkers do not verify the human implication chains, the Kang–Pikhurko theorem, or completeness of nauty's unlabelled graph generation. For $r = 8$, the independent semantic checkers establish what the retained formulas mean; the LRAT checker establishes their unsatisfiability. The human reduction is needed to connect those formulas to Theorem 5.1. For $r = 7$, the proof receipt is direct exhaustive enumeration plus an unlabelled reconstruction, not a detached certificate.

Certificate payloads are in the artifact directories named in Sections 4 and 6.

8. AI-use disclosure

OpenAI GPT 5.6 Sol was used substantially in this work. Its use included exploratory proof search, proposing intermediate lemmas, checking algebra, drafting and revising verification programs, organizing finite case reductions, preparing replay ledgers, and drafting portions of the research notes from which this manuscript was assembled. The model also helped identify failed approaches and presentation errors. This was not limited to copy editing.

OpenAI GPT-5 (Codex) was used for later source review, mathematical audits, certificate replay, and preparation of the release package. xAI Grok 4.5 was used to critique a draft.

Model output is not treated as mathematical authority. Proposed lemmas were rejected when their hypotheses could not be traced, and finite claims were kept separate from human reductions. The $r = 7$ finite lemma has two implementations with different graph-generation methods. The $r = 8$ formulas have separate production and semantic-verification paths, followed by LRAT replay. These checks reduce some risks, but they do not amount to an independent human review and do not certify the cited published theorem.

This preprint is being circulated for independent checking and has not yet received external mathematical review. The author remains responsible for the statements, proof, code, citations, and any errors.

Any circulated revision should retain a clear disclosure of the model's substantial role and link to the human-readable proof, semantic verifiers, manifests, and certificate replay instructions. The disclosure should not be read as a claim that the model independently proved or formally verified the results.

9. Nonclaims and remaining work

This preprint makes none of the following claims.

- N1. It does not solve Erdős Problem 617 for arbitrary r.
- N2. It does not settle the fixed case $r = 9$ or any larger unresolved fixed parameter.
- N3. It does not prove the cited Kang–Pikhurko theorem from first principles.
- N4. It is not a Lean formalization.
- N5. It has not been externally reviewed or accepted.
- N6. A passing finite replay does not replace the human reductions that establish completeness of the case splits.

The older degree-nine Q_3, $\mathsf{P}(8,4)$, and $\mathsf{P}(8,5)$ statements remain open as standalone claims. The fixed $r = 8$ bridge bypasses them rather than proving them.

Appendix A. Dependency ledger

result	human dependency	finite dependency
Theorem 3.1	shared colored ladder; star-and-cover propagation; three-block contradiction	weighted light-edge enumeration and unlabelled reconstruction
Theorem 5.1	shared colored ladder; 23-vertex equality-core proof; human branches of $P_3(24)$; four-block contradiction	one endpoint formula and 861 core–shell formulas, semantic reconstruction, 862 LRAT replays

The external mathematical inputs are Brooks's theorem, the Andrásfai–Erdős–Sós minimum-degree theorem, and the Kang–Pikhurko extremal theorem, including its equality description in the instances explicitly cited above.

The accompanying README identifies the source, code, proof artifacts, and replay procedure.

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